

EURSKEM AI · ENGINEERING DOCUMENTATION

VOLUME 1

Architecture Overview

How the platform is built: one typed workflow runtime, three product surfaces, and the infrastructure beneath.

Platform: Agentic Workflow Platform

Version: 0.1.0 · Generated: 2026-08-22

Built by Ayush Khandelwal

Typed workflows compiled to LangGraph · zero-token preflight · strict evidence lifecycle

1. What the platform is

Eurskem AI is an agentic workflow platform for high-stakes work. A validated YAML contract compiles to a LangGraph state graph; every node passes through one runtime boundary that binds cost context, publishes lifecycle events, writes audit records, checks for cooperative pause, resolves configuration templates, validates output against a schema, and checkpoints for recovery. The result is visible automation with typed steps, explicit evidence, bounded provider use, and human control at material decisions.

Three product surfaces share one run: the **Guided Run** for domain experts, the **Builder** for workflow authors, and the **Cockpit** for technical review. There is no second, simplified engine behind the business view — the workflow identity, run ID, node outputs, approvals and durable history are identical across all three.

2. System Architecture

Figure 1 shows the four layers: the React SPA surfaces, the FastAPI application (API, security, workflow runtime, LLM gateway), the state and storage tier, and the sidecar/integration tier, with observability spanning all of them.

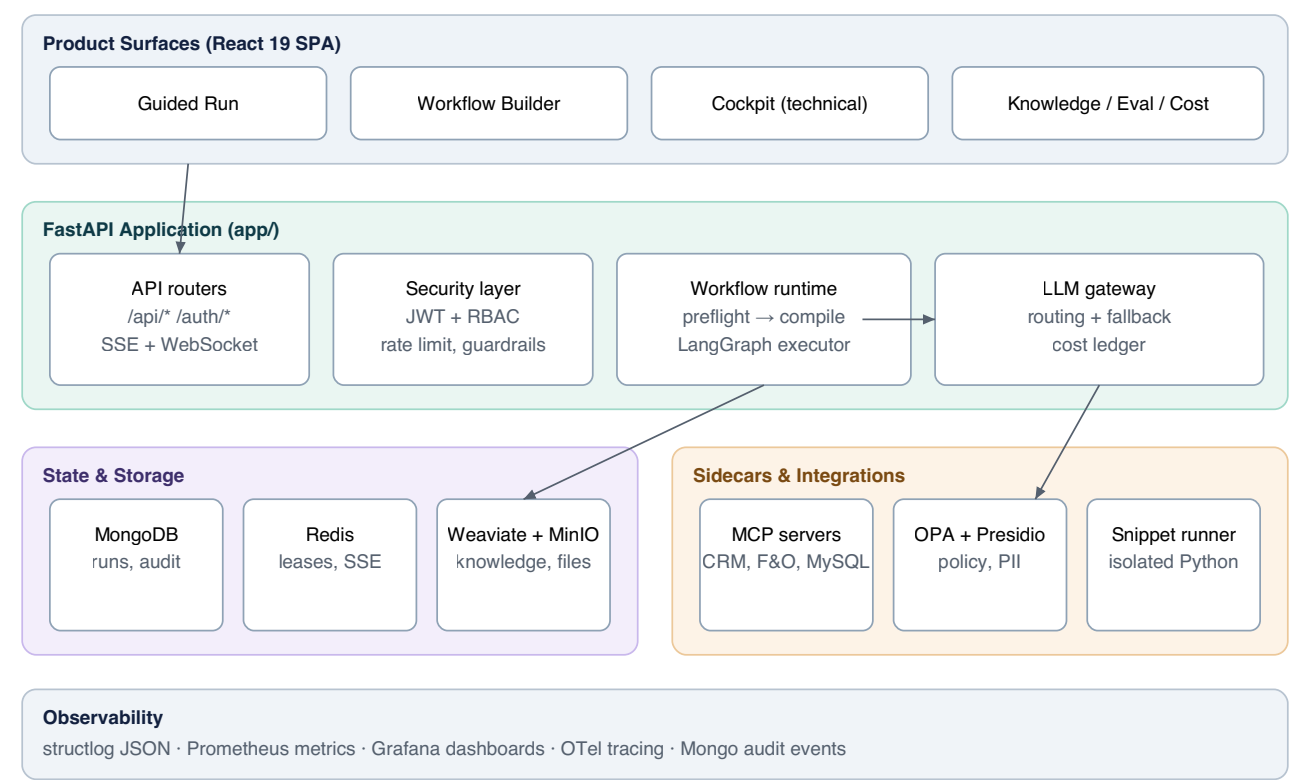
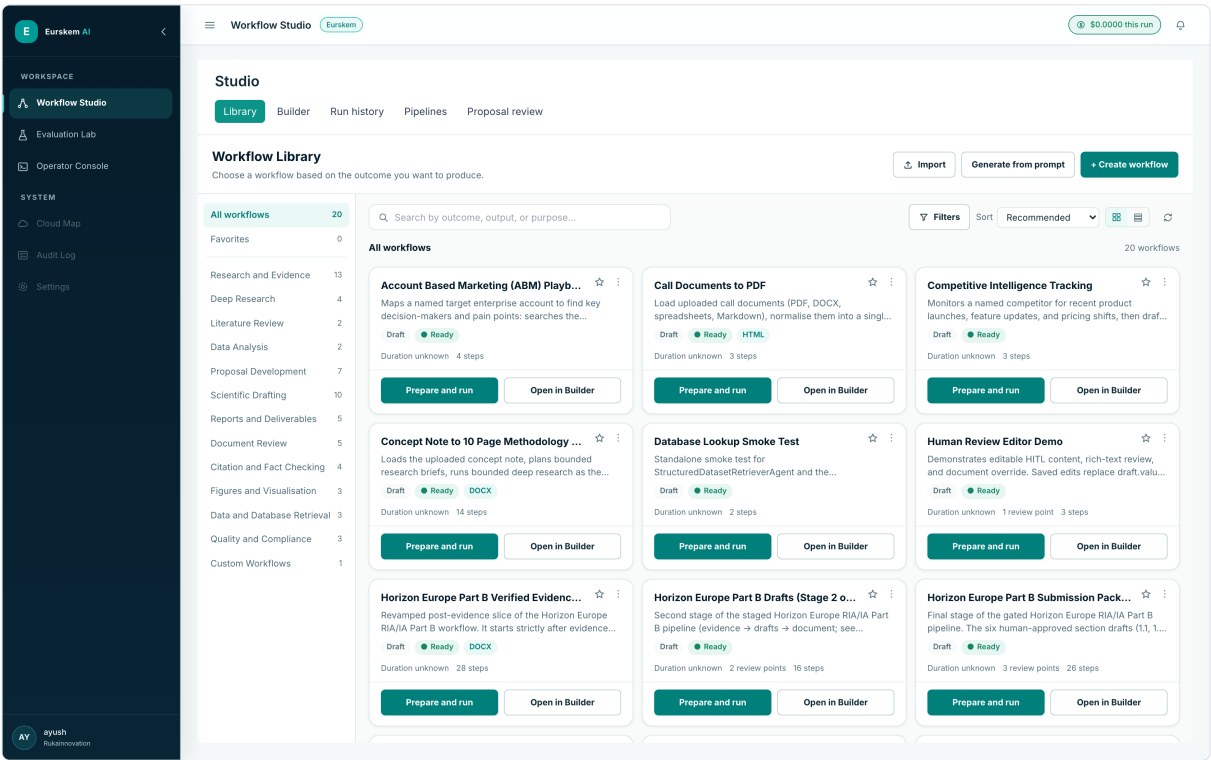


Figure 1 — System architecture.

3. The three surfaces

3.1 Workflow Library & Guided Run

Workflows are presented by outcome, not by graph shape. A run becomes business stages with plain-language explanation, one consolidated attention queue, and the deliverables.



The Workflow Library.

E

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WORKSPACE

Workflow Studio

Evaluation Lab

Operator Console

SYSTEM

Cloud Map

Audit Log

Settings

ayush

Rukminnovation

Workflow Studio

Eurskem

\$5.6495 this run

Studio

LibraryBuilderRun historyPipelinesProposal review

COMPLETED1 item need review

ViewProject AdministratorReview outputs

Concept Note to 10-Page Methodology Section

Loads the uploaded concept note, plans bounded research briefs, runs bounded deep research as the only evidence source, drafts a ~10-page methodology section in five parallel chunks, deterministically assembles it, generates figures for any image markers, and renders an editable Horizon-style DOCX. Fully autonomous - no human gates.

5 of 5 stages complete23 min 33 sec elapsedCurrent stage: Finalise

PrepareCompletedUnderstandCompletedGatherCompletedCreateCompletedFinaliseCompleted

OverviewOutputs (2)ActivityProcess map

CURRENT WORK

Render docx

Render docx finished and its contribution is available.

Why this mattersPackage the approved work and prepare the final deliverables.

Expected handoffContributes to the final deliverable and its supporting files.

Finalise milestone2 of 2 activities finished

COMPLETED WORK

Contributions to the final result

Show all stages

CONTRIBUTION READY

Graphnormalizer 1

Produced 3 structured fields.

NEXT USEContributes to the final deliverable and its supporting files.

QUALITY AND GAPSAvailable for the next stage; no separate quality result was reported.

ATTENTION

What needs you

1

Render docxReview is required before this result is ready to use.

LATEST RESULT

Outputs

View all

DOCXDOCX deliverableReady to open or downloadDownload

HTMLHTML deliverableReady to open or downloadDownload

Advanced details

Guided Run overview.

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OverviewOutputs (2)ActivityProcess map

PERSISTENT DELIVERABLES

Outputs and versions

DOCXDOCX deliverableProduced by Render docx.Download

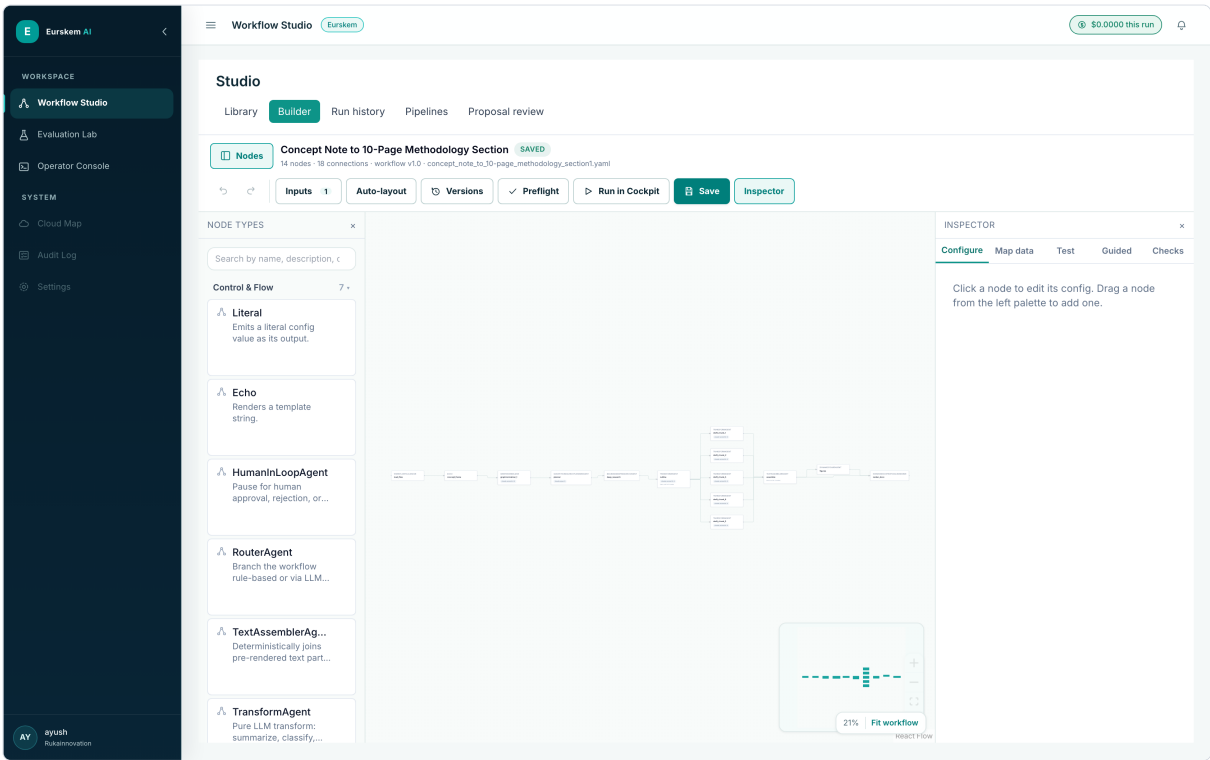
HTMLHTML deliverableProduced by Render docx.Download

Open detailed output workspace

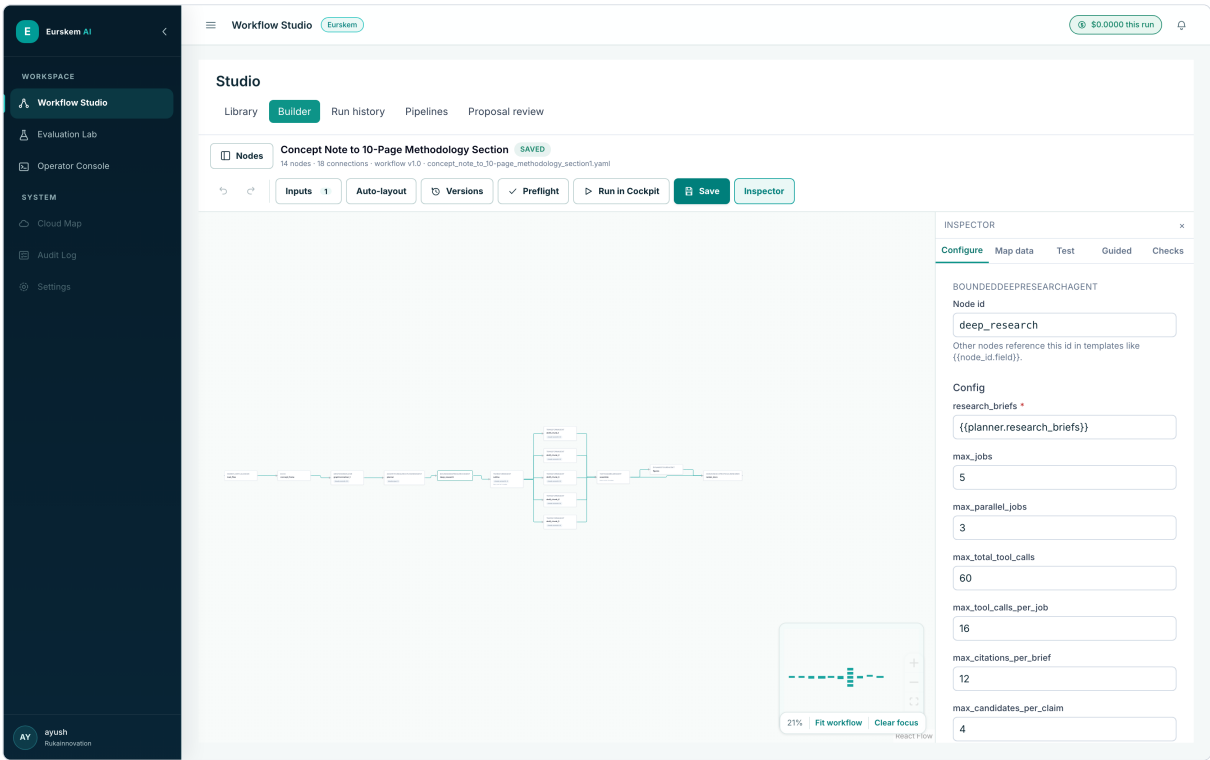
Guided Run deliverables.

3.2 Workflow Builder

A stable four-area layout: action bar, registry-driven node library, canvas, and persistent inspector. Typed configuration forms, visual data mapping, node/branch tests, autosave drafts, and immutable versions.



Builder canvas and node palette.



Typed configuration in the inspector.

The screenshot shows the Workflow Studio Builder interface. The main workspace displays a workflow graph with nodes and connections. On the right, the Inspector panel is open, showing the 'Map data' tab. The 'Visual data mapping' section explains that users can connect an available value to a text-based configuration field. The 'SELECTED NODE' is 'deep_research'. The 'Create mapping' section shows a 'Configuration field' dropdown and an 'Available value' dropdown. The 'Variable picker' shows a search bar and a list of available variables, including 'files' and 'load_files.text'.

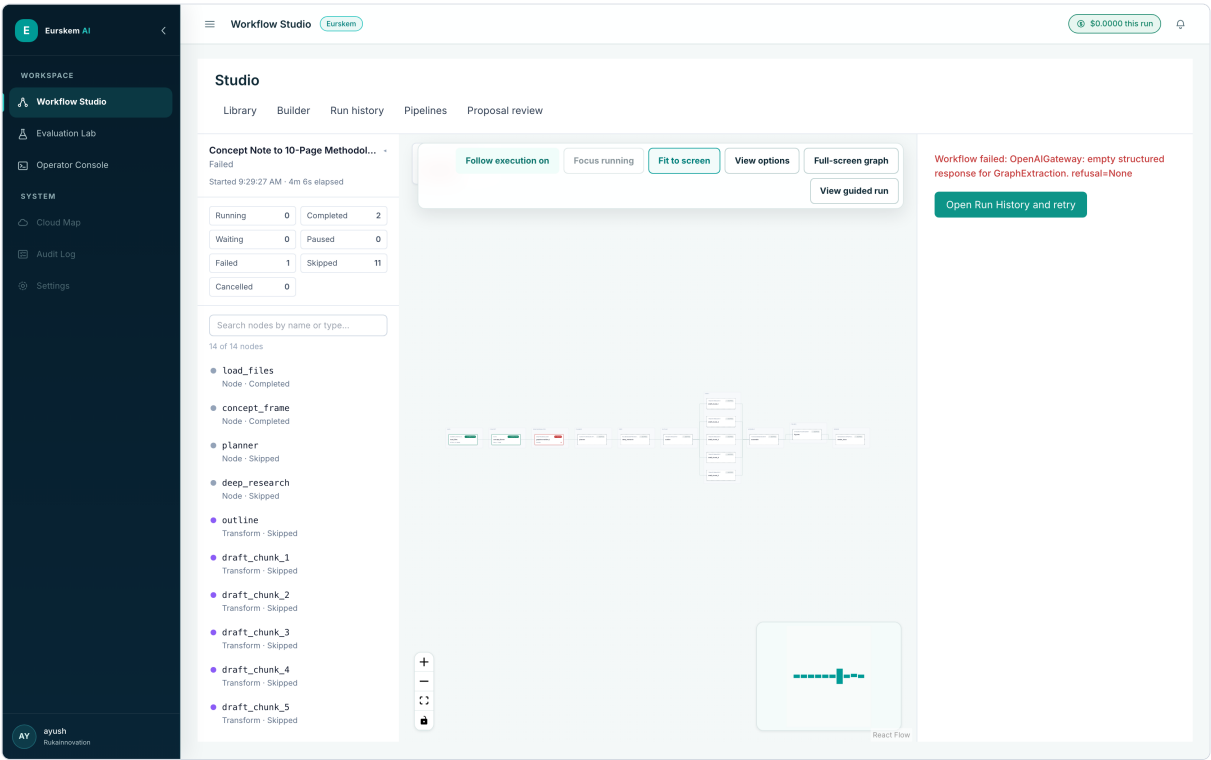
Visual data mapping between nodes.

The screenshot shows the Workflow Studio Builder interface with the 'Preflight' tab selected in the Inspector panel. The 'Preflight' section shows a 'Run preflight' button and a 'Preflight passed' status. The 'Checks' section lists various validation checks, including 'yaml', 'workflow_schema', 'node_registry', 'node_configs_and_models', 'graph_topology', 'templates', 'run_inputs', 'guided_experience', and 'graph_compile'. The 'run_inputs' check shows a warning: 'No run inputs were supplied for structural validation'.

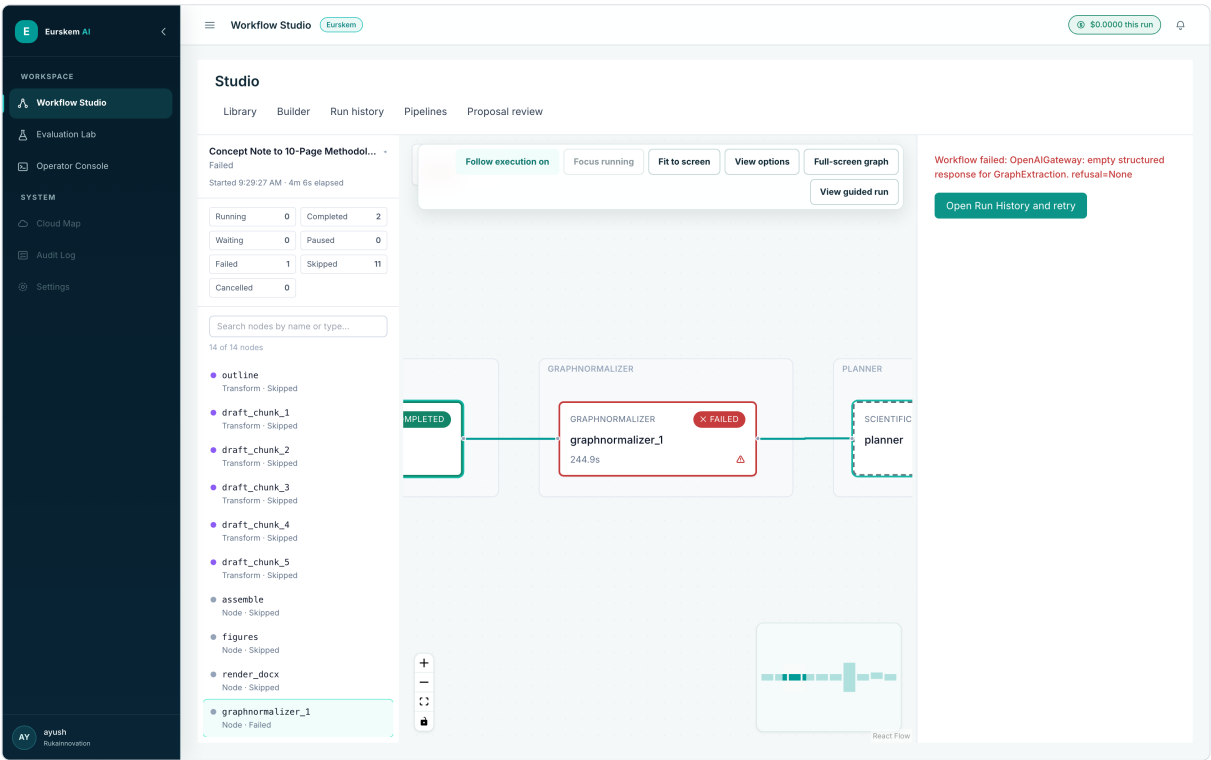
Zero-token preflight in the Builder.

3.3 Cockpit

Live graph and node lifecycle, events, failure diagnosis, audit and output inspection for technical review.



Cockpit live graph.



Node lifecycle detail.

E

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View guided run

Back

Concept Note to 10-Page Methodology Section

COMPLETEDNEEDS INPUT

Template: horizon_part_b.docx - approximately 25 pages - 81 KB - DOCX

Rendered proposal is 25 pages; configured limit is 10 pages.

DOCX81 KB ~25 pagesDownload

DOCX documents can't preview in the browser.

Download DOCX - 81 KB - ~25 pages

VariablesSources (0)Audit (14)Score

load_files

(derived from node_outputs)

concept_frame

(derived from node_outputs)

graphnormalizer_1

(derived from node_outputs)

planner

(derived from node_outputs)

deep_research

(derived from node_outputs)

outline

(derived from node_outputs)

draft_chunk_1

(derived from node_outputs)

draft_chunk_2

(derived from node_outputs)

draft_chunk_3

(derived from node_outputs)

draft_chunk_4

(derived from node_outputs)

Audit trail.

8 / 11

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4. How a workflow runs

The lifecycle of a single run is the backbone of the system:

Stage	What happens
Author	A YAML contract is written in the Builder or imported; the schema loader validates structure.
Preflight	A zero-token static pass checks topology, templates, model prerequisites and service availability — before any provider call is made.
Launch	A durable run record and checkpoint are created, then execution is detached under a Redis run-ownership lease so a duplicate launch is a no-op.
Execute	The compiled LangGraph runs; every node passes through the runtime boundary (events, cost, audit, validation, checkpoint).
Pause	Human-in-the-loop gates interrupt durably; state is persisted so resume works from any worker.
Resume	A decision rebuilds from the Redis checkpointer, replays completed nodes without re-calling providers, and continues.
Finalize	The terminal state is persisted, the projected output is computed, and pipeline/subprocess callbacks reconcile.

The screenshot displays the Eurskem AI Workflow Studio interface. The left sidebar shows the 'Workspace' with 'Workflow Studio' selected. The main area is divided into 'Run History' and 'Timeline' views. The 'Run History' view shows a list of runs with status indicators (SUCCESSFUL, FAILED) and a 'Filter' button. The 'Timeline' view shows a detailed log of nodes executed, including their status, duration, and actor. A 'Concept Note to 10-Page Methodology Section' run is highlighted, showing a successful status. The 'Timeline' view for this run shows a sequence of nodes: 'load_file', 'concept_', 'graphnor_', 'planner', 'deep_res_', 'outline', 'draft_ch_', 'draft_ch_', 'draft_ch_', 'assemble', 'figures', and 'render_d_'. The 'Inputs received' section shows the workflow variables, including 'inputs.files' and 'inputs.load_files'. The 'Outputs received' section shows the workflow outputs, including 'load_files' and 'concept_frame'.

Run timeline in Run History.

5. Evidence integrity & cost

Every call through the LLM gateway is written to the cost ledger with run ID, node ID, tokens, and both the intended and the actual model — because a fallback may execute on provider failure, and that substitution is a recorded fact rather than an invisible one. Claims are verified against sources, and the evidence lifecycle keeps provenance explicit.

The screenshot displays the Eurskem AI Workflow Studio interface. On the left is a dark sidebar with navigation options: Workspace (Workflow Studio, Evaluation Lab, Operator Console), System (Cloud Map, Audit Log, Settings), and a user profile for 'ayush'. The main panel is titled 'Studio' and shows a 'Run History' table with columns for search, filter, and status. The table lists several runs, including 'Concept Note to 10-Page Methodology Section' which is marked as 'FAILED'. A detailed view of this failed run is shown on the right, titled 'Horizon Europe Part B - Evidence (Stage 1 of 3)'. It indicates that the workflow failed at the 'evidence' stage. A red box contains the error message: 'Failed Stage: evidence. The worker process stopped unexpectedly, or the node itself raised an error. The run completed 23 of 34 nodes. Completed outputs are still available for inspection below. Retry will reuse 15 completed nodes and resume from the failed section.' Below this, a summary shows 15 completed nodes, 0 running, 1 failed, and 18 skipped. The 'Latest meaningful event' is 'Node synthesize_literature completed'. On the far right, a panel shows 'Inputs received' and 'Reusable workflow variables'.

Verified claims with evidence.

The screenshot displays the Eurskem AI Workflow Studio interface for a completed workflow. The main panel is titled 'Studio' and shows a 'Run History' table. The table lists several runs, including 'Concept Note to 10-Page Methodology Section' which is marked as 'COMPLETED'. A detailed view of this completed run is shown on the right, titled 'Concept Note to 10-Page Methodology Section'. It indicates that the workflow is 'COMPLETED' and 'NEEDS INPUT'. A 'Download' button is visible. Below this, a summary shows 'DOCX' documents, '81 KB', and '~25 pages'. The 'Latest meaningful event' is 'Node synthesize_literature completed'. On the far right, a panel shows 'Variables', 'Sources (0)', 'Audit (14)', and 'Score'. A text box contains the instruction: 'The judge scores this output against its sources. Paste an ideal proposal below to score completeness against it (optional, never stored). Without it, faithfulness and citation accuracy still apply.' Below this, a 'Score this output' button is visible.

6. Deployment & operations

Production runs on a single IONOS VPS behind Caddy (automatic TLS): an immutable release archive is transferred over SSH, validated with `sha256sum`, built with Docker Compose, health-checked through `/ready`, and automatically rolled back to the previous release if readiness fails. A post-deploy smoke/load test confirms the deployment before the `current` symlink moves. Backups capture Mongo, MinIO objects, Weaviate and Redis volumes with checksums. Containers run hardened: pinned images, read-only root filesystems, `cap_drop: ALL`, `no-new-privileges`, non-root users, and least-privilege service credentials.